

- 6 The circuit is used to supply energy to the battery from the generator. A variable resistor of resistance R is used to control the current in a circuit, as shown in Fig. 6.1.

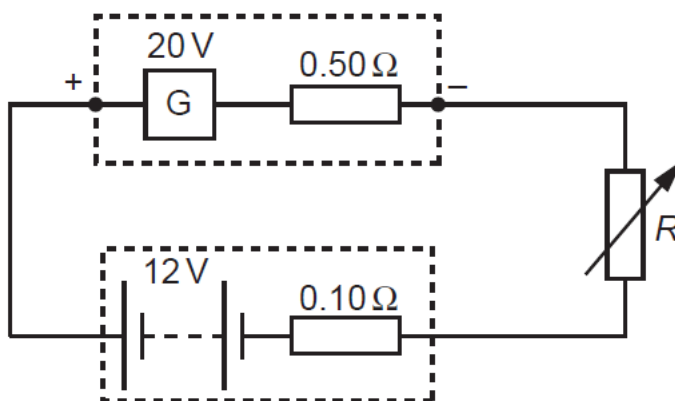


Fig. 6.1

The generator G has e.m.f. 20 V and internal resistance $0.50\ \Omega$. The battery has e.m.f. 12 V and internal resistance $0.10\ \Omega$. The current in the circuit is 2.0 A.

- (a) Determine the resistance R .

resistance $R = \dots\dots\dots\ \Omega$ [2]

$$\Sigma E = \Sigma IR$$

$$20 - 12 = 2.0(0.6 + R) \quad \text{C1}$$

$$R = 3.4\ \Omega \quad \text{A1}$$

- (b) Calculate the total power generated by G

power = $\dots\dots\dots$ W [2]

$$P = EI \quad \text{C1}$$

$$= 20 \times 2$$

$$= 40\ \text{W} \quad \text{A1}$$

- (c) Calculate the power loss in the total resistance of the circuit.

Power loss = $\dots\dots\dots$ W [2]

$$P = I^2 R \quad \text{C1}$$

		$P = (2)2 \times (0.1 + 0.5 + 3.4)$ $= 16 \text{ W}$	A1
	(d)	Determine the efficiency of the circuit.	
		efficiency =	[3]
		$\text{efficiency} = \text{useful power} / \text{output power}$ $= 24 / 40$ $= 60\%$	M1 C1 A1
	(e)	A student suggests that the value of resistance R will not affect the efficiency calculated in (d). Explain if you agree with the student.	
			
			
			[1]
		Agree, since efficiency = $V_{\text{battery}}/E_{\text{generator}}$ which is independent of R.	B1
			[Total: 10]

7	(a)	A spring that has an unstretched length of 0.500 m is attached to a fixed point. A mass of 0.400
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